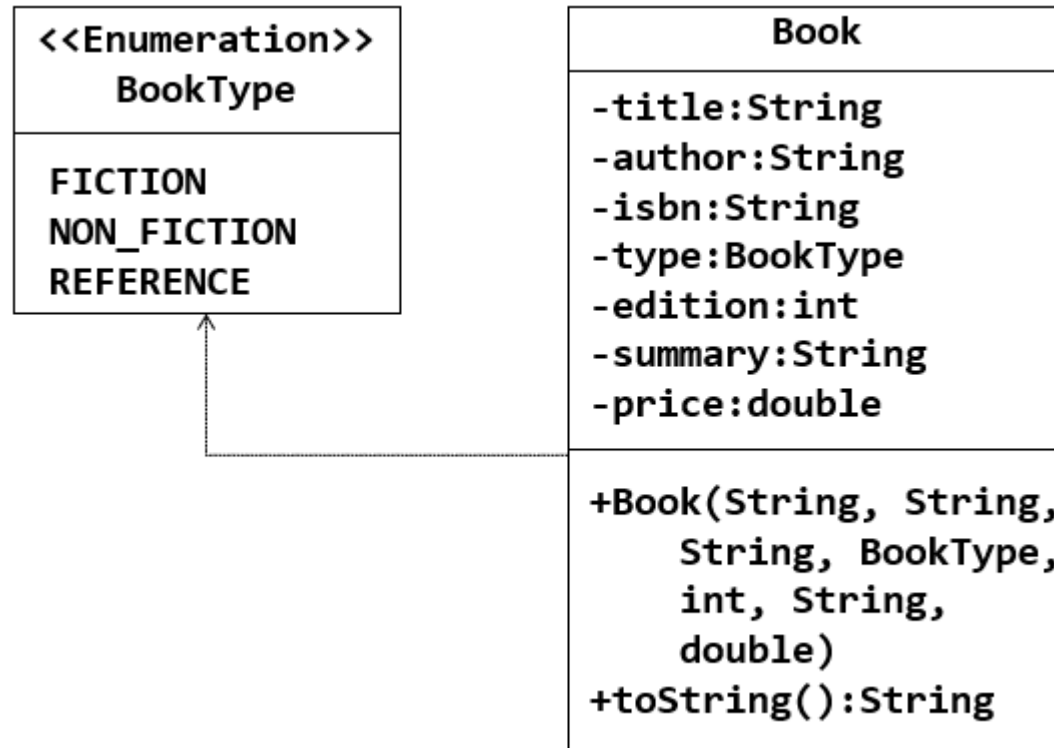


CSC1029 OO Programming

Lecture 17 – Review of Assessment 1



Assessment 1 Outline Solution (1)



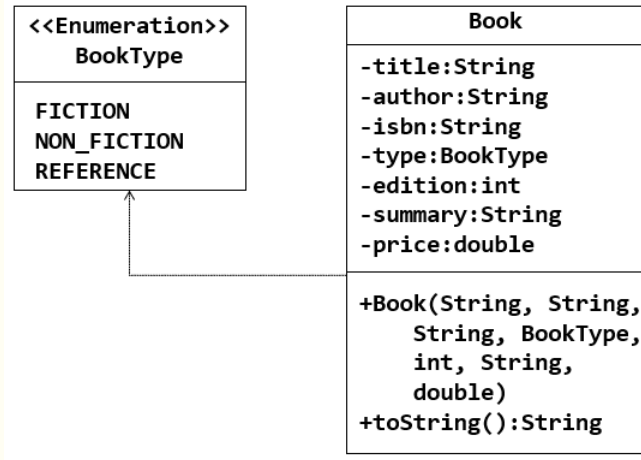
Part 1: Enum & Book Class

BookType
Book
BookTester

Part 2: Multiple Book Instances

addStockItem()
displayItems()

Assessment 1 Outline Solution (2)



```
public enum BookType {
    FICTION("Stories from the imagination"),
    NON_FICTION("Factual, real-life experiences"),
    REFERENCE("Theories, Philosophies and common knowledge");

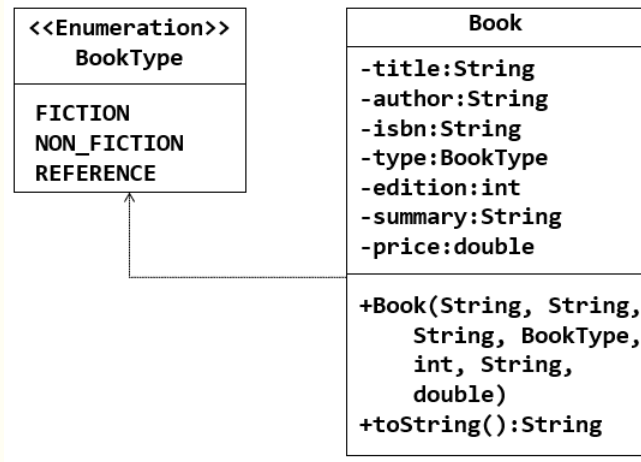
    private String info;

    private BookType(String str) {
        info = str;
    }

    public String toString() {
        return info;
    }
}
```

Part 1: Enumeration

Assessment 1 Outline Solution (3)

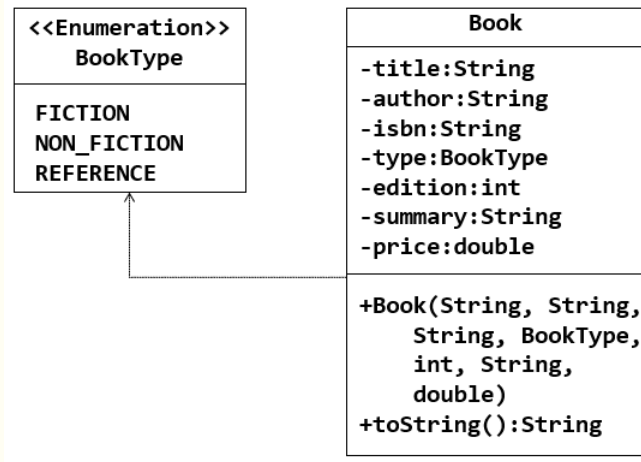


```
public class Book {
    private String title; // book title
    private String author; // book author
    private String isbn; // book isbn number
    private BookType type; // type of book
    private int edition; // book edition
    private String summary; // book summary
    private double price; // price of book
}
```

*Part 1: Book
(attributes)*

}

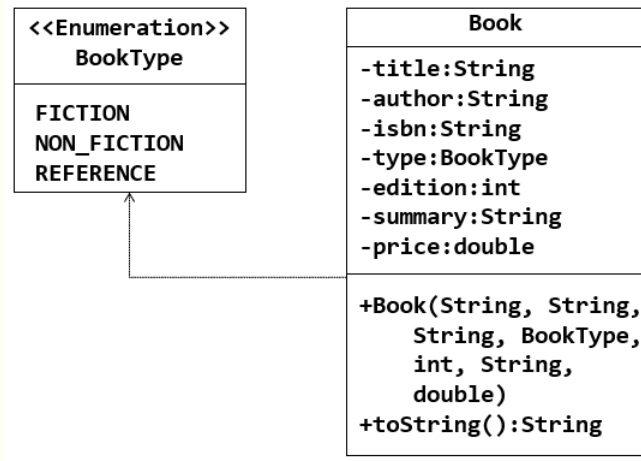
Assessment 1 Outline Solution (4)



```
public Book(String title, String author, String isbn,
            BookType bType, int edition, String summary,
            double price) {
    this.title = title;
    this.author = author;
    this.isbn = isbn;
    this.type = bType;
    this.edition = edition;
    this.summary = summary;
    this.price = price;
}
```

*Part 1: Book
(constructor)*

Assessment 1 Outline Solution (5)



Part 1: Book (getters)

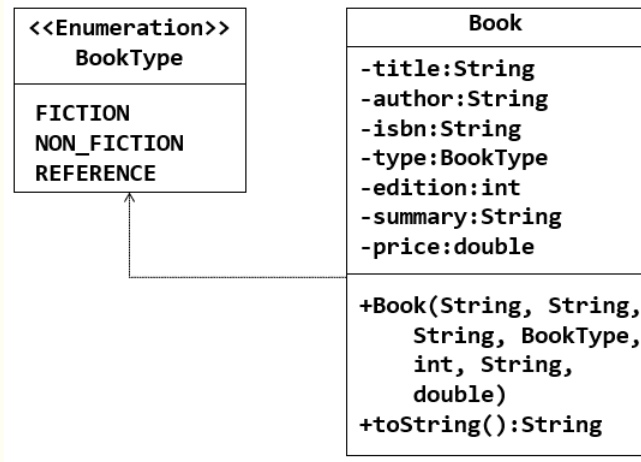
```
public String getTitle() {  
    return this.title;  
}
```

```
public String getAuthor() {  
    return this.author;  
}
```

```
public String getIsbn() {  
    return this.isbn;  
}
```

```
public BookType getType() {  
    return this.type;  
}
```

Assessment 1 Outline Solution (6)



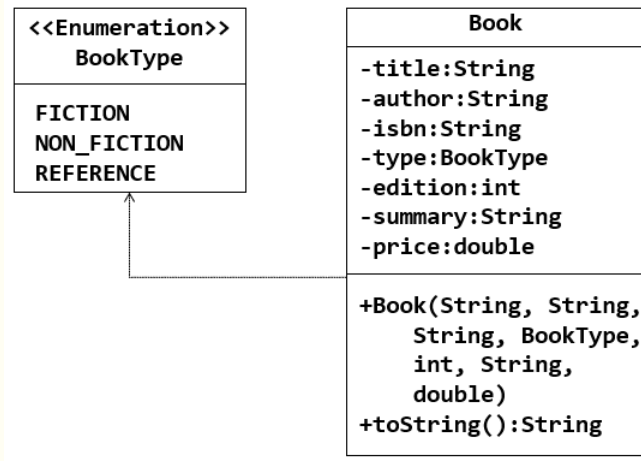
```
public int getEdition() {  
    return this.edition;  
}
```

```
public String getSummary() {  
    return this.summary;  
}
```

```
public double getPrice() {  
    return this.price;  
}
```

*Part 1: Book
(getters)*

Assessment 1 Outline Solution (7)

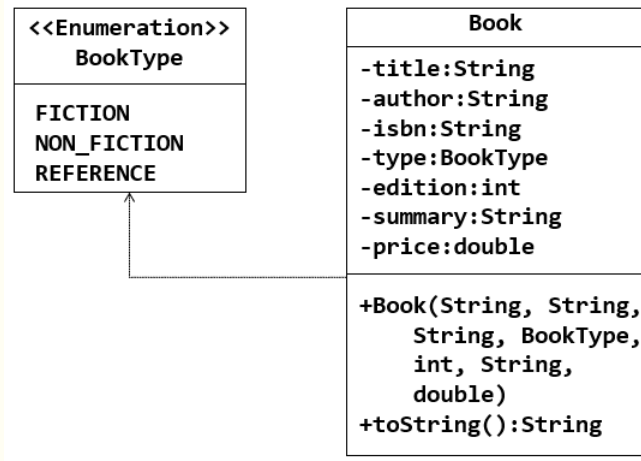


```
public String toString() {
    String res = "";
    res += this.title + " by " + this.author + "\n";
    res += "ISBN: " + this.getIsbn() + "\n";
    res += "Type: " + this.getType() + "\n";
    res += "Edition: " + this.getEdition() + "\n";
    res += "Summary: " + this.getSummary() + "\n";
    res += "Price: " + String.format("£%.2f",
                                    this.getPrice()) + "\n";

    return res;
}
```

Part 1: Book
(toString)

Assessment 1 Outline Solution (8)



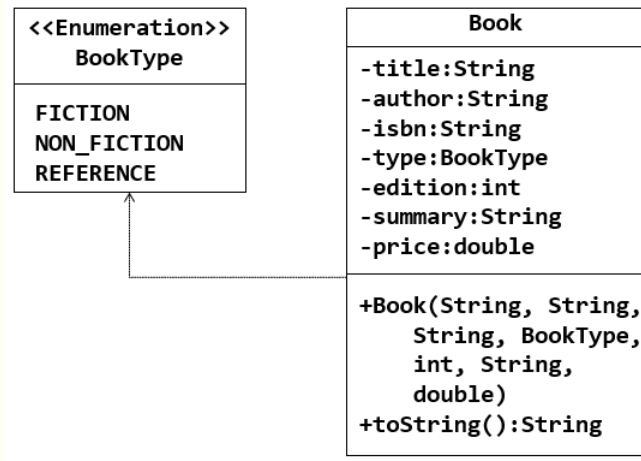
```
public static void main(String[] args) {
    Book b1 = new Book("The Hunger Games", "Suzanne
        Collins", "140713208", BookType.FICTION, 1,
        "First in the ground-breaking HUNGER GAMES
        trilogy", 8.90);

    Book b2 = new Book("Java, the Complete Reference",
        "Herbert Schildt", "126046341",
        BookType.REFERENCE, 12,
        "The definitive guide to Java programming",
        32.20);

    System.out.println(b1);
    System.out.println(b2);
}
```

Part 1: BookTester

Assessment 1 Outline Solution (9)



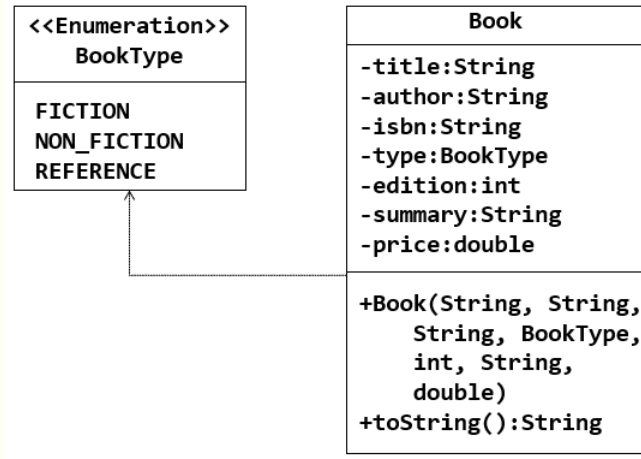
```
Book b1 = new Book("The Hunger Games", "Suzanne  
Collins", "140713208", BookType.FICTION, 1,  
"First in the ground-breaking HUNGER GAMES  
trilogy", 8.90);
```

```
Book b2 = new Book("Java, the Complete  
Reference", "Herbert Schildt", "126046341",  
BookType.REFERENCE, 12,  
"The definitive guide to Java programming",  
32.20);
```

Part 2: BookTester (ArrayList)

```
// ArrayList declaration & addition of  
instances  
ArrayList<Book> data = new ArrayList<Book>();  
data.add(b1);  
data.add(b2);
```

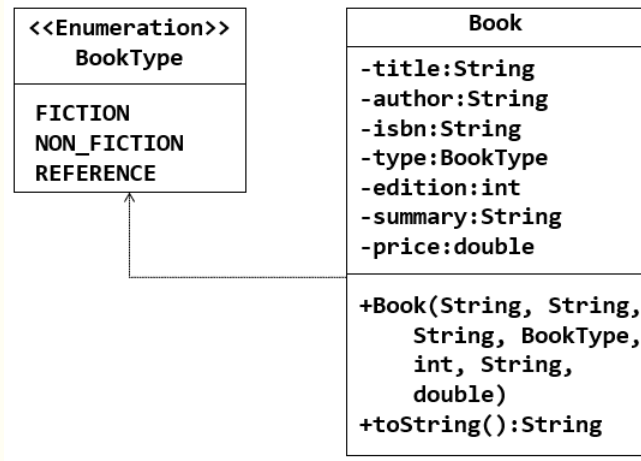
Assessment 1 Outline Solution (10)



```
public static void displaySummary(ArrayList<Book> books) {
    if (books == null || books.size() == 0) {
        System.out.println("No Books Available.");
    } else {
        for (Book bk : books) {
            System.out.println(bk.getTitle() + ", " +
                               bk.getSummary());
        }
    }
}
```

Part 2: BookTester

Assessment 1 Outline Solution (11)



```
System.out.println("Testing displaySummary");
System.out.println("1. When books are available in
the list ...");
displaySummary(data); // should display all books
```

```
System.out.println("2. When list is null ...");
displaySummary(null); // null Array List
```

```
System.out.println("3. When list is empty ...");
displaySummary(new ArrayList<Book>()); // an empty
Array List
```

Part 2: BookTester

Assessment 2 Making a Start ⁽¹⁾

Start Now! (do not delay this)

1. Implement class Book (start with A1 version)
2. Implement the outline structure of **LibraryBook**
3. Implement the outline structure **Library**
4. Implement the outline structure of the menu app

You will need searching & sorting for some
Features – to be introduced next week.



What's Next?

Searching Data

More examples on ...

inheritance

interfaces

enumerations

object references

